L D College of Engineering
Computer Engineering Department

BE05000461 – Fuzzy Logic and Genetic Programming
Problem Based Learning (PBL):
Design and Development of a Fuzzy Controller for Real-World Engineering Problem

1. Background

Many real-world problems involve uncertainty, vagueness, and approximate reasoning. Traditional Boolean logic often fails to model such situations effectively. Fuzzy Logic provides a mathematical framework to represent human reasoning and make intelligent decisions.

In this PBL, you have to identify a real-world engineering problem from your discipline (recommended otherwise you may choose any real-world problem) and develop a fuzzy logic controller or decision-support system.

2. Objectives

After completing this PBL, students will be able to:

- Identify a real-world problem suitable for fuzzy logic.
- Define linguistic variables and membership functions.
- Design fuzzy IF–THEN rules.
- Implement a Mamdani Fuzzy Inference System using Python (scikit-fuzzy).
- Test and evaluate the controller.
- Present the developed solution.

3. Learning Outcomes

After completing this PBL, students should be able to:

- Model vague real-world problems using fuzzy logic.
- Design membership functions and linguistic variables.
- Construct an effective fuzzy rule base.
- Implement a complete Mamdani fuzzy inference system in Python.
- Analyze controller performance through testing.
- Communicate engineering solutions through reports and presentations.

4. Team Formation

- **Group Size:** 3–4 students
- Each group must select one unique domain/application.
- No two groups should work on the same problem.
- After forming team and problem identification, you need to submit details via google form which will be shared to you.

5. Suggested Domains (Not Limited to)

Agriculture

- Smart Irrigation Controller
- Fertilizer Recommendation
- Pest Risk Assessment
- Crop Disease Prediction
- Greenhouse Climate Control

Healthcare

- Diabetes Risk Assessment
- Heart Disease Risk Prediction
- ICU Patient Monitoring
- Hospital Bed Priority
- Mental Stress Evaluation

Smart Cities

- Smart Traffic Signal Controller
- Smart Parking Recommendation
- Street Light Brightness Control
- Water Distribution Control
- Waste Collection Scheduling

Automobile

- Automatic Braking System
- Adaptive Cruise Control
- Fuel Efficiency Advisor
- Driver Fatigue Detection
- Parking Assistance

Consumer Electronics

- Washing Machine Controller
- Air Conditioner Temperature Control

- Refrigerator Cooling Controller
- Camera Autofocus
- Smart Fan Speed Control

Environmental Engineering

- Air Quality Index Assessment
- Flood Risk Prediction
- Water Quality Evaluation
- Forest Fire Risk Assessment
- Noise Pollution Analysis

Industrial Automation

- Boiler Temperature Control
- Robot Speed Controller
- Conveyor Belt Speed Control
- Production Scheduling
- Machine Health Monitoring

Education

- Student Performance Evaluation
- Placement Eligibility Prediction
- Scholarship Recommendation
- Attendance Monitoring
- Course Recommendation

6. Expected Workflow

- Phase 1 – Problem Identification – Identify the domain of your branch.
- Phase 2 – System Design –
    - Identify inputs, outputs
    - Range of input and output
    - Identify linguistic variables of each input/output and their corresponding values.
    - Decide the membership function (Triangular, Trapezoidal, Gaussian)
    - Construct Rule Base – Identify at least 6-8 rules.
    - Take one sample data, calculate memberships, Identify which rules are fired and their Firing strength, Defuzzify it using weighted average and generate the crisp output.
- Phase 3 – Python Implementation – Use scikit-fuzzy library
- Phase 4 – Testing - Test at least 10 different input combinations
- Phase 5 – Analysis – Discuss controller behaviour, strengths and weaknesses, Scope of improvement
- Phase 6 – UI – Convert your fuzzy controller into a nice user interface.

7. Milestones and Deliverables:

Milestones:

- **Milestone-1 (Deadline-16.08.2026):** Design of a Fuzzy Controller as described in Phase-2 of a previous section. (Upload ppt/docx/Handwritten System Design PDF)
- **Milestone-2 (Deadline-23.08.2026):** Implementation of a Fuzzy Controller in Python as described in Phase-3 of a previous section. (Upload .ipynb file)
- **Milestone-3 (Deadline-30.08.2026):** Testing, Analysis and UI as described in Phase-4,5 and 6 in previous section. (Upload .ipynb file and simulator screenshots)

Deliverables After Project Completion (Deadline: 13.09.2026):

- **Presentation**: 10–12 slides (Problem, Objectives, System Design, Rules, Implementation, Results, Analysis)
- **Report**: Well formatted (Problem, Objectives, System Design, Rules, Implementation, Results, Analysis)
- **Python Source Code:** .ipynb or .py file
- **Create GitHub repo**
- **Make short video and upload on YouTube/Insta/LinkedIn**
- **Make a LinkedIn Post**

Note: Presentations of this PBL project will commence after your Mid-Semester Exam.